Vaibhav Panchal

AI/ML Developer | GenAI & LLM Systems Engineer

+91 88662 31808 | vaibhavpanchal1808@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI/ML Developer at TestGrid Labs Inc. building intelligent test automation pipelines using deep learning and LLMs. M.E. in Software Engineering (LDCE, GTU, CGPA 8.21/10). Research work in Computer Vision (YOLOv11 + ECA, +3.8% mAP50 on DIOR-R, under review) and Cybersecurity (hybrid IDS, 98.4% accuracy, unpublished manuscript). Hands-on across the full ML lifecycle from data engineering and model training to Flask deployment and CI/CD integration. Core stack: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain, Flask, Nginx, AWS, Docker.

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, JavaScript, C/C++, Java, HTML/CSS

Frameworks and Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, XGBoost, LangChain, LangGraph, FastAPI, Flask, Streamlit

AI and ML: Computer Vision, NLP, LLMs, RAG, Object Detection, Prompt Engineering, OpenAI API, Ollama, RAGAS, Langfuse, NeMo Guardrails, OCR, ONNX, VLMs

Cloud and Infrastructure: AWS (EC2), Docker, Nginx

Databases: SQLite, MongoDB, ChromaDB, Pinecone (Vector DB)

Tools and Practices: Git, Jupyter, MLflow, Postman, Selenium Grid, Bitbucket, REST API Design, CI/CD

WORK EXPERIENCE

Junior AI/ML Developer | *TestGrid Labs Inc.* | *Navsari, Gujarat* May 2026 - Present

- Designed and built an in-house vision model pipeline (VLM, ONNX-based icon naming, two-pass OCR) served through a FastAPI application, replacing a third-party LLM API dependency and improving detection of missing UI elements.
- Optimized an image-based element detection feature, cutting processing time by ~85% and fixing missing-element defects reported by users.
- Researched evaluation frameworks for AI agents, built test case collections to validate new agents, and tested new AI features through REST APIs with Postman. Explored local-model deployment to reduce reliance on external APIs.
- Built backend tooling for a secure tunneling product: server-side log replication for faster debugging, usage tracking, and overload warnings. Delivered client-specific fixes and releases.

Trainee Associate Software Developer | *Techreale Pvt. Ltd.* | *Valsad, Gujarat* Jan 2024 - Apr 2024

- Delivered production features for the Famyll healthcare platform (React.js, Node.js, MongoDB) across claim management, auth flows, and real-time communication modules.
- Reduced API response handling bugs by refactoring async integration patterns; contributed across 3 sprint cycles in an Agile team of 8.

Summer Intern, Frontend Development (React) | *INFOLABZ IT Services Pvt. Ltd.* | *Ahmedabad, Gujarat* Jul 2023

- Developed and optimized React.js dashboard components with async API integration; improved rendering performance through targeted component-level refactoring.

PROJECTS

MESH Support - Multi-Agent Customer Support Platform 2026

Python | LangGraph | FastAPI | MongoDB | GPT-OSS-120B | NeMo Guardrails | Langfuse | Pytest

- Architected MESH Support using LangGraph to orchestrate a Supervisor Agent that classifies and routes queries to specialized Billing, Technical, and Refund sub-agents, each with isolated prompts, tools, and memory, backed by FastAPI, MongoDB, and Groq's GPT-OSS-120B.
- Integrated NVIDIA NeMo Guardrails input/output rails to block jailbreak, prompt-injection, off-topic, and unsafe messages, achieving 100% precision/recall (0% false-positive rate) on a labeled safety set with only ~1.8s added overhead per turn.
- Instrumented every turn with Langfuse distributed tracing and shipped a real-time observability dashboard visualizing agent hops and tool calls; verified reliability with an automated pytest suite (21/21 tests passing).
- Delivered 100% end-to-end accuracy on the evaluation suite (routing, tool-selection, and guardrail decisions) with a ~5.4s median guarded response time, validated by automated end-to-end test and evaluation harnesses (23/23 backend tests passing).

RAG System with Evaluation Pipeline 2026

Python | FastAPI | React | Pinecone | Langfuse | RAGAS | MiniLM | BM25 | Groq (gpt-oss-120b) | MongoDB

- Built hybrid retrieval using Pinecone (MiniLM 384-d) + in-memory BM25 with Reciprocal Rank Fusion (k=60, α=0.6), enabling keyword-only fallback when the vector store is unavailable.

- Created a 45-question RAGAS golden set covering faithfulness, answer relevancy, and context precision, with a FastAPI evaluation endpoint and React dashboard backed by MongoDB.
- Added pre-retrieval prompt-injection/jailbreak filtering and routed out-of-scope queries through RAG for grounded refusals instead of canned responses.
- Instrumented retrieval and generation with Langfuse, and deployed with API auth, upload limits, non-root containers, and CI checks for linting, compilation, and integration tests.

ExamForge - LLM-Powered Exam Question Generator 2026

Python | LangChain | Ollama | OpenAI GPT | Streamlit

- Built an end-to-end RAG pipeline: ingests syllabus PDFs, chunks and embeds content, then prompts an LLM to generate structured question papers with configurable difficulty and Bloom's taxonomy tagging.
- Supported both local inference (Ollama) and cloud (OpenAI API), making it deployable in air-gapped institutional environments; cut manual question-drafting time by ~80% in pilot use.
- Engineered structured JSON output parsing with validation to ensure generated questions conform to a strict schema before rendering to PDF.

Content-Based Movie Recommendation Engine Jan - Feb 2025

Python | Scikit-learn | TF-IDF | Cosine Similarity | Streamlit

- Implemented TF-IDF vectorization and cosine similarity on the TMDB 5000 dataset; deployed as a real-time Streamlit app with sub-second recommendation latency.
- Pipeline is architected for extensibility - collaborative filtering and hybrid approaches can be plugged in without restructuring the serving layer.

RESEARCH WORK

Channel Attention for Oriented Object Detection in Remote Sensing 2025 (Under Review)

YOLOv11-OBB | ECA Attention | DIOR-R Dataset | PyTorch

- Integrated Efficient Channel Attention (ECA) into YOLOv11s-OBB to improve feature recalibration for small and densely packed objects in aerial imagery.
- Evaluated on DIOR-R (20-class remote sensing benchmark): achieved +3.8% mAP50 and +4.3% recall over the baseline, with no increase in parameter count.

Hybrid Deep Learning Architecture for Network Intrusion Detection 2025 (Manuscript in Preparation)

CNN + BiLSTM + Attention | NSL-KDD | CICIDS | TensorFlow

- Proposed a sequential CNN-BiLSTM-Attention IDS that captures both spatial and temporal patterns in network flow data; applied PCA and AutoEncoder for feature compression.
- Achieved 98.4%+ accuracy on NSL-KDD and CICIDS; outperformed Random Forest and SVM across all attack categories on precision, recall, and F1-score.

EDUCATION

M.E. - Software Engineering - GTU, LDCE, Ahmedabad 2024-2026 | CGPA: 8.21/10

B.E. - Computer Engineering - GTU, GDEC, Navsari 2020-2024 | CGPA: 8.11/10

ACHIEVEMENTS & CERTIFICATIONS

- Evaluate Generative AI Applications - Microsoft Learn; covered evaluation frameworks and metrics for Azure, Foundry Tools, and Content Safety. (2026)
- Microsoft Learn Challenge | Ignite Edition: Build Trustworthy AI Solutions on Microsoft Azure. (2025)
- SSIP Hackathon 2022 - Developed an automated smart attendance management system; selected among top teams from Gujarat. (2022)
- Dev-o-lution Hackathon - Certificate of Participation for building a competitive solution under a 24-hour constraint. (2025)
- Introduction to Model Context Protocol – Anthropic. (2026)
- Model Context Protocol: Advanced Topics – Anthropic. (2026)

LEADERSHIP & EXTRACURRICULAR

- Volunteered at college-level tech events and coding quiz competitions, supporting event coordination and participant engagement.
- Academic Mentorship - Mentored B.E. students in computer science subjects during my M.E., providing guidance on technical concepts and problem-solving.

INTERESTS

AI/ML Research, LLMs & RAG Systems, Computer Vision, Deep Learning, MLOps, Automation, Cloud Computing, Data Science | Chess, Exploring Emerging Technologies, Music, Video Games